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Case 36-2025: A 55-Year-Old Woman with Dyspnea, Fatigue, and Gastrointestinal Bleeding

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PRESENTATION OF CASE

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Dr. Olivia Kahn-Boesel (Medicine): A 55-year-old woman was admitted to this hospital because of recurrent rectal bleeding and shortness of breath.

The patient had been in her usual state of health until 6 months before the current presentation, when hematochezia developed. She was evaluated at the emergency department of another hospital. The hemoglobin level was 8.0 g per deciliter (reference range, 12.0 to 16.0), with a normal mean corpuscular volume; the white-cell count and platelet count were normal. The patient was admitted to the hospital, and colonoscopy was performed; internal hemorrhoids were detected, but no masses, ulcers, or polyps were noted. Treatment with oral iron was initiated, and she was discharged home.

Three months before the current presentation, hematochezia developed again, and the patient presented to the emergency department of a second hospital. The hemoglobin level was 6.4 g per deciliter, with a mean corpuscular volume of 73 fl (reference range, 80 to 100). The white-cell count was 3700 per microliter (reference range, 4000 to 11,000), and the platelet count was normal. The ferritin level was 5 μ g per liter (reference range, 10 to 200), and the transferrin saturation was 3% (reference range, 14 to 50). The patient was admitted to the second hospital. Two units of packed red cells were transfused, and an intravenous infusion of iron was administered.

Esophagogastroduodenoscopy (EGD) with additional enteroscopy reportedly showed a normal-appearing esophagus, stomach, and duodenum. Owing to the patient's anatomy, only the proximal jejunum could be seen and its appearance was normal; no blood was noted in the gastrointestinal tract. The rectal bleeding abated spontaneously, and the patient was discharged home.

Two weeks later, the patient was evaluated in the outpatient gastroenterology clinic of the second hospital. Video-capsule endoscopy was reportedly negative. Given the negative endoscopic evaluation, treatment with scheduled iron infusions was started.

Two months before the current presentation, the patient had recurrent hematochezia, along with fatigue, prompting her to present to the emergency department of this

hospital. The vital signs were normal, and the hemoglobin level was 6.8 g per deciliter, which led to admission to this hospital. Results of serum protein electrophoresis were normal, as were the serum levels of free kappa and free lambda light chains. The blood levels of copper, lead, and zinc were also normal. One unit of packed red cells was transfused, and intravenous iron was administered, which resulted in a decrease in symptoms. Given the recent negative endoscopic evaluation and the absence of ongoing rectal bleeding, further investigation was deferred, and the patient was discharged home with instructions to return to her gastroenterologist for follow-up care.

On the day of the current presentation, melena developed, and the patient returned to the emergency department of this hospital. On presentation, the vital signs were normal. The hemoglobin level was 6.5 g per deciliter, the white-cell count

3170 per microliter with a normal differential count, and the platelet count 144,000 per microliter.

Dr. Robert G. Rasmussen: Computed tomography (CT) with angiography was performed according to a gastrointestinal bleeding protocol that was specific to the abdomen and pelvis. No active gastrointestinal bleeding was identified on imaging. The celiac and mesenteric arterial vasculature was normal, including a patent hepatic artery and superior mesenteric artery (Fig. 1A and 1B). There was occlusion of the left portal vein with restoration of flow in the distal segment (Fig. 1C). The right portal vein was dilated, and stenosis was present at the junction of the right portal vein and the main portal vein, which was patent (Fig. 1D and 1E). Associated splenomegaly was noted, along with portosystemic collaterals in the hepatic hilum. The study also showed evidence of a previous hepaticojunostomy in the hepatic hilum

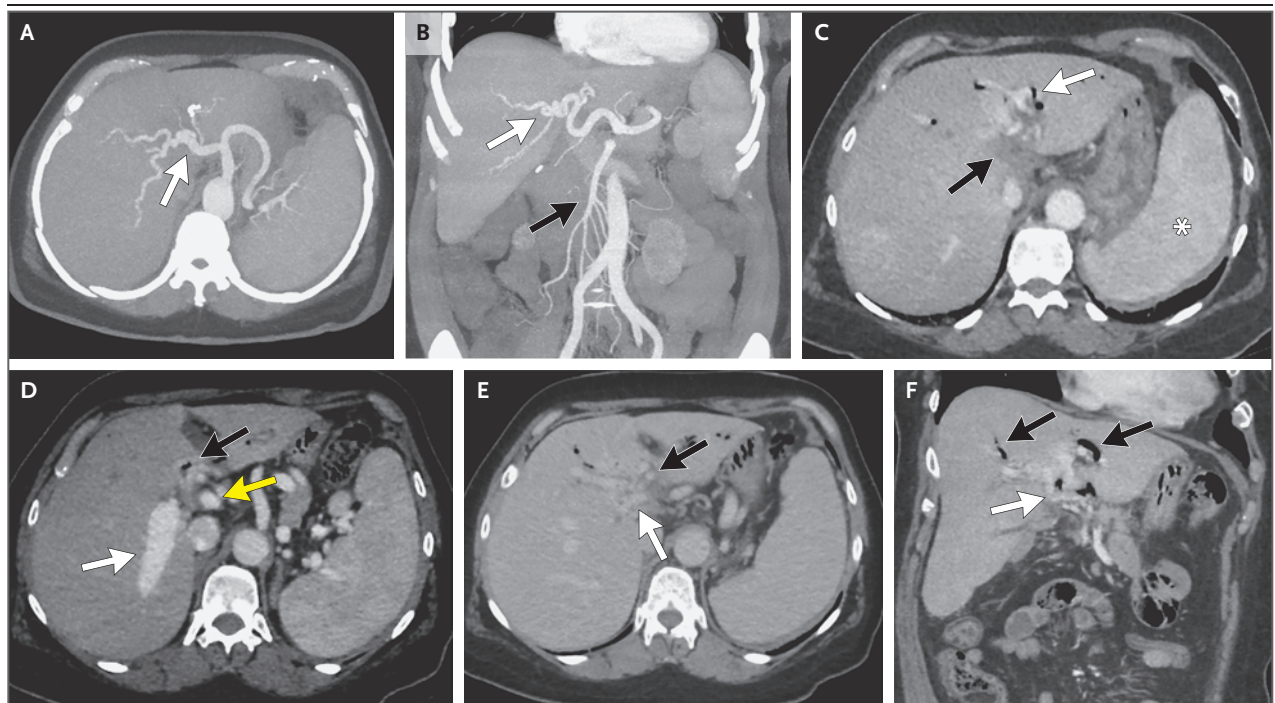


Figure 1. CT Angiography.

CT angiography was performed to evaluate for active gastrointestinal bleeding. An axial maximum-intensity-projection (MIP) image obtained in the early arterial phase (Panel A) shows a patent hepatic artery (arrow) at the level of the celiac axis. A coronal MIP image obtained in the early arterial phase (Panel B) shows the patent hepatic artery (white arrow) and a patent superior mesenteric artery (black arrow). An axial image obtained in the late arterial phase (Panel C) shows occlusion of the left portal vein (black arrow) with distal reconstitution (white arrow) and splenomegaly (asterisk). A more inferior axial image (Panel D) shows a dilated right portal vein (white arrow), a patent main portal vein (black arrow), and portosystemic collaterals in the hepatic hilum (yellow arrow). A superior axial image (Panel E) shows stenosis at the junction of the right portal vein and the main portal vein (white arrow) and additional portosystemic collaterals (black arrow). A coronal image at the level of the hepatic hilum (Panel F) shows evidence of a previous hepaticojunostomy surrounded by portosystemic collaterals (white arrow) and the presence of expected pneumobilia in the left and right hepatic lobes (black arrows).

surrounded by the collaterals and the presence of expected pneumobilia associated with hepaticojejunostomy (Fig. 1F).

Dr. Kahn-Boesel: One unit of packed red cells was transfused, and the patient was admitted to this hospital for management of symptomatic anemia and recurrent hemorrhage. On evaluation, she reported ongoing hematochezia and melena, as well as fatigue and mild dyspnea on exertion. A review of systems was negative for weight loss, night sweats, abnormal bleeding from other mucosal sites, bruising, jaundice, pruritus, pedal edema, abdominal pain, and distention.

The patient's medical history was notable for seronegative ocular myasthenia gravis and symptomatic premature ventricular complexes, for which she had undergone cardiac ablation. Surgical history was notable for cholecystectomy, which had been performed 20 years before the current presentation; the procedure had been complicated by laceration of the bile duct, for which she had undergone open repair and hepaticojejunostomy. Appendectomy had been performed 4 years before the current presentation. The patient took no medications. She did not take nonsteroidal antiinflammatory drugs or dietary supplements. She had no known adverse reactions to medications. She lived in Massachusetts with her husband and adult child. She did not smoke, drink alcohol, or use recreational drugs. She was a retired educator. She reported no recent travel and had not lived outside the United States. She had no family history of liver disease, bleeding disorders, or cancer.

The oral temperature was 36.1°C, the blood pressure 99/55 mm Hg, the heart rate 61 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 99% while the patient was breathing ambient air. The weight was 85 kg, and the body-mass index (the weight in kilograms divided by the square of the height in meters) was 30. On examination, the skin was pale but not jaundiced. The abdomen was soft and nontender, without distention or palpable organomegaly. She did not have Terry's nails, spider angiomas, or palmar erythema.

The results of kidney-function tests were normal, as were blood levels of aspartate aminotransferase, alanine aminotransferase, bilirubin, and albumin. The prothrombin time was 13.9 seconds (reference range, 10.0 to 13.0), and the international normalized ratio was 1.2 (reference range,

0.8 to 1.1). Additional laboratory test results are shown in Table 1.

Dr. Rasmussen: Magnetic resonance cholangiopancreatography (MRCP) revealed an atrophic caudate lobe of the liver with associated focal intrahepatic biliary ductal dilatation (Fig. 2A and 2B). In addition, diffusion-weighted images showed a region of abnormal increased signal intensity (Fig. 2C and 2D) and abnormal increased delayed enhancement (Fig. 2E and 2F) that extended from the perihilar region to the superior surface of the liver near the confluence of the hepatic veins. These findings were associated with attenuation of the left and middle hepatic veins (Fig. 2E), occlusion of the main left portal vein, and stenosis at the origin of the right portal vein.

Dr. Kahn-Boesel: EGD and colonoscopy showed erythematous mucosa and erosive gastropathy in the gastric antrum, as well as a hyperplastic polyp. Diverticulosis was also detected, without any vascular lesions or active bleeding.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Irun Bhan: This 55-year-old woman with a history of complicated cholecystectomy presents after having four episodes of gastrointestinal bleeding in a span of 6 months. The first step in the management of acute gastrointestinal bleeding is stabilization through volume resuscitation. Fortunately, the patient's condition remained hemodynamically stable during these episodes. Nonetheless, her hemoglobin level, which had initially been normal, decreased over time, and iron-deficiency anemia developed — a finding that indicated clinically significant blood loss. I will approach this patient's case by attempting to establish the anatomical location of the source of bleeding and the mechanistic cause of her bleeding.

DETERMINING THE SOURCE OF BLOOD LOSS

After initial stabilization, the next step in the management of gastrointestinal bleeding is to determine the source of blood loss. The color of the stool can provide insight into the general location of gastrointestinal bleeding. This patient initially presented with hematochezia, the passage of red blood in the stool, which is indicative of either a lower gastrointestinal hemorrhage or a brisk episode of upper gastrointestinal bleeding. The subsequent occurrence of rectal bleeding that was

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	2 Mo before Current Presentation	On Current Presentation
Hemoglobin (g/dl)	12.0–16.0	6.8	6.5
Hematocrit (%)	41.0–53.0	23.5	32.2
White-cell count (per μ l)	4500–11,000	5490	3170
Differential count (per μ l)			
Absolute neutrophils	1800–7700	3460	1610
Absolute lymphocytes	1000–4800	1510	1270
Absolute monocytes	200–1200	390	190
Absolute eosinophils	0–900	90	70
Absolute basophils	0–300	20	20
Immature granulocytes	0–100	0	0
Platelet count (per μ l)	150,000–400,000	167,000	144,000
Mean corpuscular volume (fl)	80–100	83	86
Sodium (mmol/liter)	135–145	141	141
Potassium (mmol/liter)	3.4–5.0	4.1	3.7
Chloride (mmol/liter)	98–108	108	108
Carbon dioxide (mmol/liter)	23–32	23	25
Urea nitrogen (mg/dl)	8–25	12	17
Creatinine (mg/dl)	0.60–1.50	0.68	0.60
Calcium (mg/dl)	8.5–10.5	8.7	9.0
Glucose (mg/dl)	70–110	86	108
Aspartate aminotransferase (U/liter)	10–40	22	27
Alanine aminotransferase (U/liter)	10–55	33	30
Alkaline phosphatase (U/liter)	15–115	91	83
Total bilirubin (mg/dl)	0.0–1.2	0.3	0.2
Albumin (g/dl)	3.3–5.0	4.2	3.8
Globulin (g/dl)	1.9–4.1	2.7	2.6
Lead (μ g/dl)	0.0–3.5	<1.0	—
Copper (μ g/dl)	77–206	142	—
Zinc (μ g/dl)	60–106	62	—
IgG (mg/dl)	614–1295	828	—
IgA (mg/dl)	69–309	207	—
IgM (mg/dl)	53–334	78	—
Free kappa light chain (mg/liter)	3.3–19.4	15.4	—
Free lambda light chain (mg/liter)	5.7–26.3	14.3	—
Kappa:lambda ratio	0.3–1.7	1.08	—
Serum protein electrophoresis	Negative for monoclonal component	Negative for monoclonal component	—
Prothrombin time (sec)	9.4–12.5	14.8	13.9
International normalized ratio	0.8–1.1	1.3	1.2

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

described as melena is consistent with an upper gastrointestinal source, although bleeding in the proximal colon can also cause melena. Since this patient had passed both stool containing red blood and melena, the location of the bleeding is most likely in the upper gastrointestinal tract. Furthermore, the presence of both gastrointestinal hemorrhage and thrombocytopenia prompts consideration of underlying portal hypertension. In the context of portal hypertension, thrombocytopenia can occur as a result of splenic sequestration of platelets¹ and, in persons with chronic liver disease, decreased thrombopoietin production.^{2,3} In this patient, the finding of thrombocytopenia is particularly notable because reactive thrombocytosis often occurs in the context of iron-deficiency anemia.⁴

OBSCURE GASTROINTESTINAL BLEEDING

The patient underwent EGD twice and also had a colonoscopy. On the first EGD, which was per-

formed with the use of enteroscopy, the small bowel could be seen through the first portion of the jejunum. After the first EGD, the patient underwent video-capsule endoscopy, which is designed to examine the entire length of the small bowel. Despite the use of these procedures, no upper gastrointestinal source of bleeding was identified. Furthermore, colonoscopy was performed to evaluate the large bowel from the rectum through the terminal ileum; a bleeding source was not identified. Thus, the patient's presentation can be classified as obscure gastrointestinal bleeding⁵ — persistent or recurrent bleeding with no source identified on upper and lower endoscopic evaluation.

In this case, we are left with two possibilities: the patient's previous evaluations missed the source of bleeding, or the source is an area of the bowel that has not yet been examined. Missing the site of blood loss is unlikely. Still, it can occur when there is no active bleeding at the time

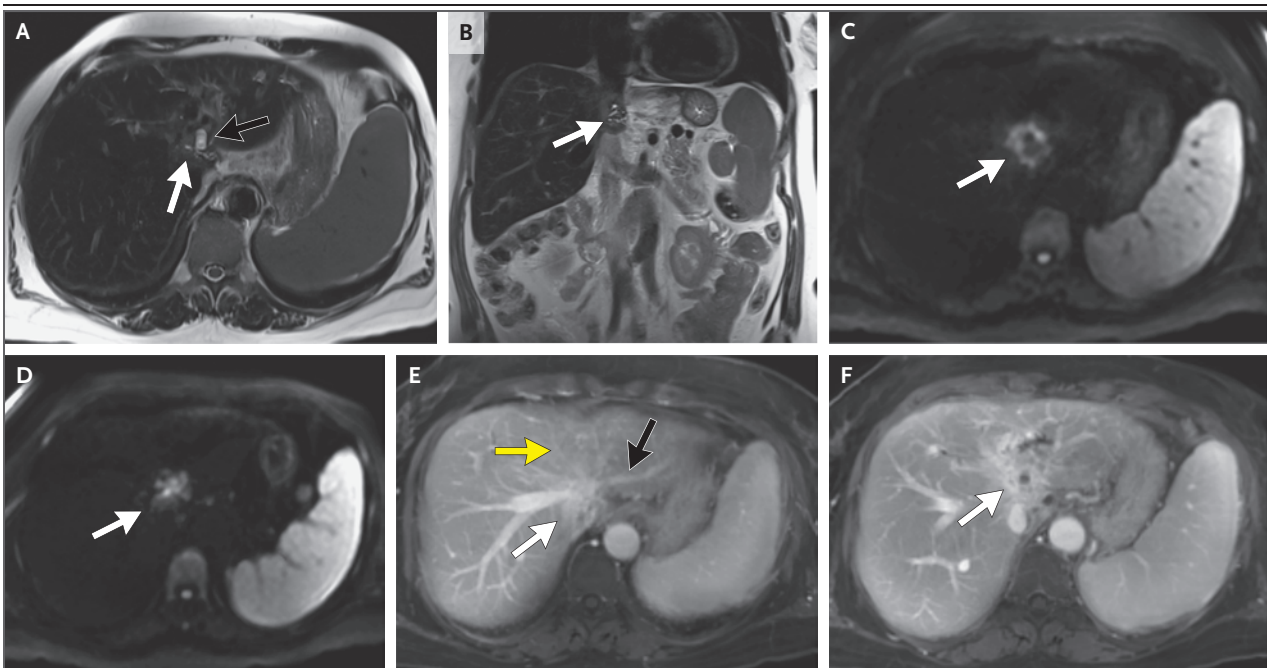


Figure 2. Magnetic Resonance Cholangiopancreatography.

An axial T2-weighted image (Panel A) shows an atrophic caudate lobe (white arrow) with focal intrahepatic biliary ductal dilatation (black arrow). A coronal T2-weighted image (Panel B) also shows focal biliary ductal dilatation and atrophy in the caudate lobe (arrow). High b-value (1400 seconds per square millimeter) diffusion-weighted images (Panels C and D) show abnormal increased signal intensity near the confluence of the hepatic veins (Panel C, arrow) and in the perihilar region (Panel D, arrow). Axial images obtained after the administration of contrast material (Panels E and F) show abnormal increased delayed enhancement near the confluence of the hepatic veins (Panel E, white arrow) and in the perihilar region (Panel F, arrow). Panel E also shows attenuation of the middle hepatic vein (yellow arrow) and left hepatic vein (black arrow).

of evaluation and when the underlying mucosal abnormality is subtle, as in the case of bleeding associated with a Dieulafoy's lesion. The possibility of bleeding from an unexamined source should be strongly considered, since this patient has altered anatomy as a result of a previous Roux-en-Y hepaticojejunostomy, and the Roux limb has not yet been observed.

MECHANISMS OF GASTROINTESTINAL BLEEDING

The patient's history and physical examination provide little additional information that would indicate a mechanism for her bleeding. Despite the presence of thrombocytopenia that suggests portal hypertension, she has no clinically significant risk factors for liver cirrhosis, such as excess alcohol consumption, metabolic risk factors (except for her body-mass index), or chronic viral infection. Furthermore, there are no physical examination findings or additional laboratory findings that are suggestive of advanced liver disease.

Although cirrhosis accounts for the majority of cases of portal hypertensive bleeding, the possibility of noncirrhotic portal hypertension is a consideration in this patient. The features that were seen on cross-sectional imaging, including occlusion of the left portal vein, stenosis of the right portal vein, splenomegaly, and portosystemic collaterals near the hepaticojejunostomy site, are all suggestive of clinically significant noncirrhotic portal hypertension. This constellation of clinical and radiologic findings prompts consideration of jejunal variceal hemorrhage in the Roux limb from the patient's hepaticojejunostomy. Although focal biliary ductal dilatation is probably an incidental finding and is most likely due to postoperative architectural changes, the unlikely possibility of hemobilia caused by a vascular–biliary fistula is a consideration.

Finally, it will be critical to ascertain the cause of the portal-vein obstruction at the hepatic hilum. A cancer in the hilum, such as cholangiocarcinoma, should be considered, especially given the presence of biliary ductal dilatation, but this possibility seems unlikely on the basis of the patient's normal bilirubin and alkaline phosphatase levels, the absence of weight loss or other constitutional symptoms, and the absence of a hilar mass or stricture on MRCP.

The patient's history of myasthenia gravis could suggest a separate, unrelated autoimmune

source of fibrosis, although this possibility is unlikely. Ultimately, the cause of the portal-vein obstruction is most consistent with postoperative changes, including fibrosis, resulting from the remote complicated cholecystectomy, as indicated by the caudate-lobe atrophy with abnormal increased signal intensity and delayed enhancement in the perihilar region. The poststenotic dilatation of the right portal vein is most likely due to turbulent blood flow passing through the stenosis. However, another contributor to this focal vascular enlargement could be an arteriovenous fistula that is causing retrograde flow through the right portal vein and contributing to portal hypertension.

The most likely cause of this patient's bleeding is a jejunal varix. This possibility could be further investigated with angiography of the portal venous system, which would confirm the finding of portosystemic collaterals near the jejunum, even if there is no active extravasation at the time of the procedure. The procedure could also be therapeutic through embolization of the culprit vessel. Furthermore, the pressure gradient across the stenosis in the right portal vein could be measured and, if physiologically significant, the stenosis could be dilated and possibly stented to improve portal outflow, improve portal pressures, and reduce the risk of recurrent portal hypertensive complications. Lastly, the unlikely possibility of a fistula between an artery and the portal vein could be ruled out with angiography during the same procedure.

The most appropriate approach to gaining access to the portal venous system is probably percutaneously into the area of poststenotic dilatation. Targeting the enlarged vessel is expected to be straightforward under ultrasound guidance. Transjugular access, although commonly performed, may involve the need to puncture through more hepatic parenchyma between the hepatic vein and right portal vein, which makes this approach more technically challenging and associated with a relatively increased risk of bleeding. A spontaneous portosystemic shunt that would permit portal venous system access through the central venous system (e.g., a splenorenal shunt accessed through the femoral vein) was not identified on cross-sectional imaging. Transsplenic access is an alternative and less commonly pursued method, and determining the safety profile of this technique warrants further examination.⁶

DR. IRUN BHAN'S DIAGNOSIS

Jejunal variceal bleeding due to noncirrhotic portal hypertension.

DIAGNOSTIC IMAGING

Dr. Vincent Wu: Given the patient's ongoing melena and hematochezia, she was referred for invasive angiography and selective portal venography. While she was under general anesthesia, mesenteric angiography was performed to evaluate for an arterial source of gastrointestinal hemorrhage. Access to the right common femoral artery was obtained, and the celiac artery was selected for evaluation; an arteriogram did not show active bleeding in the stomach or duodenum (Video 1A). In addition, no arteriportal fistula involving the

hepatic arteries was detected. Next, the superior mesenteric artery was selected for evaluation; an arteriogram of the superior mesenteric artery also did not show gastrointestinal bleeding (Video 1B). The delayed portal venous phase of the superior mesenteric arteriogram showed the jejunal varices and stenosis in the portal vein (Video 1B). An arteriogram of the inferior mesenteric artery did not show colonic or rectal bleeding (Video 1C).

Attention was turned to the portal venous system. Percutaneous transhepatic access to the right portal vein was obtained under ultrasound guidance. A vascular sheath was placed, and an initial portal venogram obtained with the use of a flush catheter showed severe stenosis of the portal vein, with contrast material filling the jejunal varices and cavernomas through the jejunal vein at the level of the hepaticojejunostomy site (Fig. 3A and

Videos showing angiography and management of bleeding are available at [NEJM.org](https://www.nejm.org)

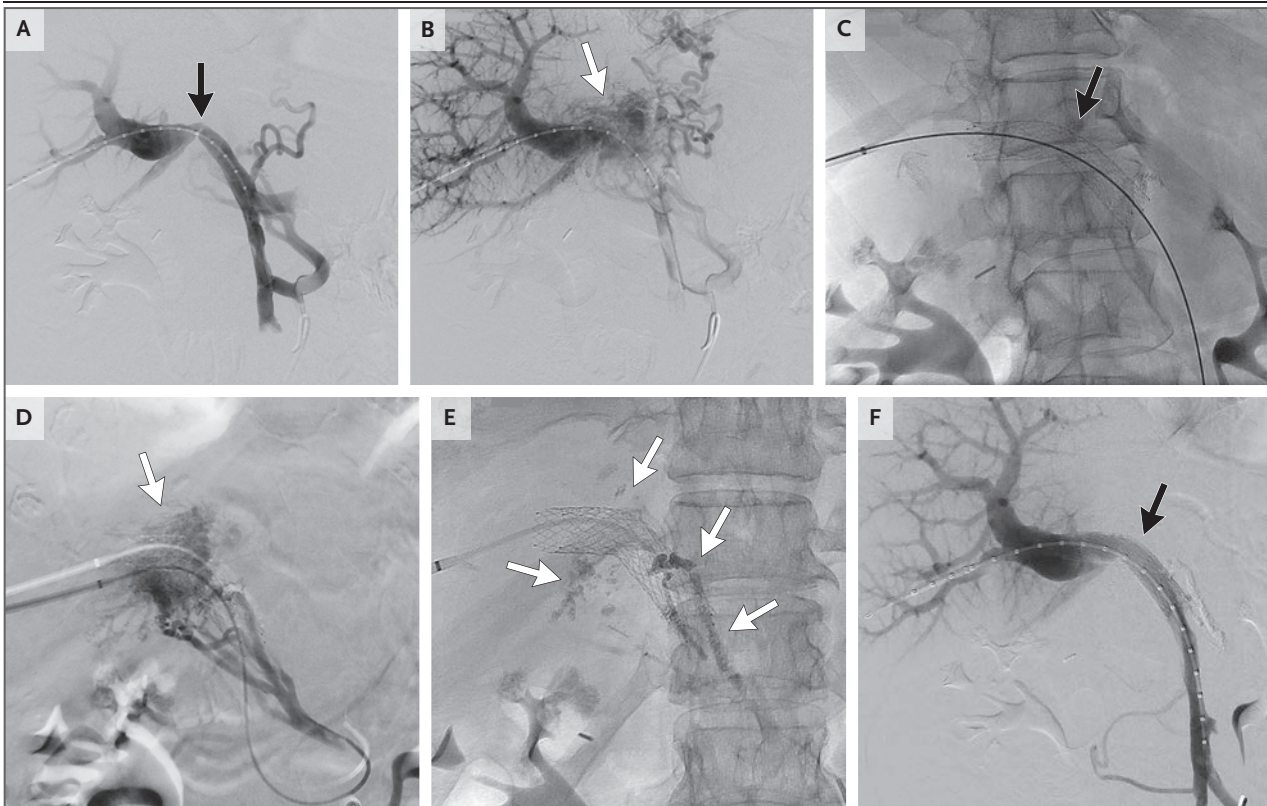


Figure 3. Interventional Radiology Imaging Studies.

An early-phase portal venogram (Panel A) shows severe stenosis of the portal vein (arrow), and a delayed-phase portal venogram (Panel B) shows filling of cavernomas and jejunal varices (arrow). A fluoroscopic image (Panel C) shows placement of a self-expanding bare-metal stent across the portal-vein stenosis (arrow). Selective venograms show the jejunal varices (Panel D, arrow), followed by embolization of the jejunal varices with the use of *N*-butyl-2-cyanoacrylate (Panel E, arrows). A postembolization portal venogram shows a patent portal-vein stent (Panel F, arrow) with in-line portal flow from the superior mesenteric vein into the intrahepatic portal veins and elimination of jejunal variceal filling after embolization.

3B and Video 1D). The pressure gradient across the stenosis was elevated, measuring 11 mm Hg (normal portal-vein pressure gradient, <5),⁷ which was suggestive of hemodynamically significant stenosis. On the basis of the portal venogram, the source of gastrointestinal hemorrhage was most likely the jejunal varices, which had formed at the level of the hepaticojejunostomy site due to portal-vein stenosis. This is an example of prehepatic, noncirrhotic portal hypertension resulting in the formation of ectopic varices that can be prone to hemorrhage.⁸ In this patient's case, portal-vein stenosis was most likely related to inflammation and adhesions from bile-duct injury and bile leak from the patient's previous cholecystectomy.

RADIOLOGIC DIAGNOSIS

Portal-vein stenosis causing postsurgical jejunal variceal bleeding at the level of the hepaticojejunostomy site.

DISCUSSION OF MANAGEMENT

Dr. Wu: Stenting of the portal-vein stenosis was performed with a 12-mm by 60-mm self-expanding bare-metal stent (Fig. 3C and Video 1E). The stent was expanded with a 12-mm angioplasty balloon. With the use of a microcatheter, the jejunal varices were selected and then embolized with a liquid embolic mixture of *N*-butyl-2-cyanoacrylate, lipiodol, and tantalum powder (Fig. 3D and 3E and Video 1F). A postembolization portal venogram showed a widely patent portal-vein stent with in-line flow of contrast material from the superior mesenteric vein into the intrahepatic portal vein (Fig. 3F and Video 1G). The filling of the jejunal varices had stopped. After stenting, the pressure gradient across the previous location of the portal-vein stenosis had decreased to 2 mm Hg. To prevent hemorrhage from the transhepatic access after removal of the vascular sheath, tract embolization was performed with coils and absorbable gelatin pledgets. Hemostasis of the right common femoral artery access was achieved with the use of a vascular closure device.

FOLLOW-UP

Dr. Wu: On the night of the stenting procedure, fever developed, which was attributed to postem-

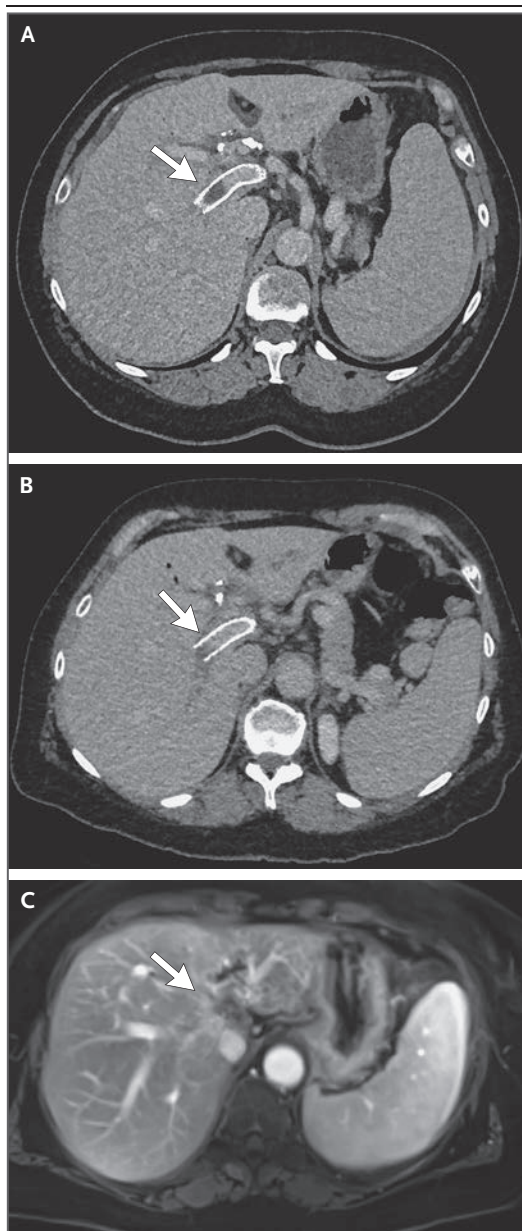


Figure 4. Follow-up Imaging Studies of the Abdomen and Pelvis.

A contrast-enhanced CT angiogram obtained 18 days after the stenting procedure (Panel A) shows a nonocclusive thrombus in the distal end of the stent (arrow). After initiation of anticoagulant therapy, a CT angiogram obtained 2 months after the stenting procedure (Panel B) shows a decrease in the extent of the thrombus (arrow). A magnetic resonance cholangiopancreatograph obtained 5 months after the stenting procedure (Panel C) shows a reduction in perihilar enhancement of the liver, which is suggestive of decreased filling of the jejunal varices and cavernomas (arrow).

bolization syndrome, given that no clinical signs of infection (other than fever) were present, no growth was seen on blood cultures, and there was no evidence of pneumonia on chest radiography. The day after the procedure, the hemoglobin level was 6.7 g per deciliter, and 1 unit of packed red cells was transfused. The hemoglobin level subsequently increased to 8.7 g per deciliter. Over the course of the next few days, the patient's melena abated. She was discharged home 4 days after the procedure.

CT angiography of the abdomen performed 18 days after the stenting procedure showed a nonocclusive thrombus in the distal end of the portal-vein stent (Fig. 4A). Thrombus formation in this location was thought to be associated with a size mismatch between the stent and the post-stenotic dilatation of the portal vein, resulting in turbulent flow. At this time, the patient's gastrointestinal bleeding had not recurred; thus, a decision was made to start treatment with apixaban for the thrombus associated with the stent.

Two months after the stenting procedure, follow-up CT angiography showed a decrease in the extent of the thrombus and a patent portal-vein stent (Fig. 4B). Four months after the procedure, the hemoglobin level was 12.4 g per deciliter. The patient was able to continue anticoagulant therapy without bothersome side effects, and she had had no further episodes of gastrointestinal bleeding. MRCP performed 5 months after the procedure (Fig. 4C) showed a radiologically significant decrease in perihilar hepatic enhancement, which indicated decreased filling of the varices near the hepaticojejunostomy site.

This finding is not consistent with a hilar neoplasm as the cause of her portal-vein occlusion.

Dr. Joshua C. Ziperstein (Medicine): Does it seem odd that this marked portal-vein bleeding commenced briskly and suddenly after so many decades if the bleeding was due to benign postsurgical changes?

Dr. Bhan: It is undoubtedly an unusual aspect of the case. This may have been a situation in which years of remodeling in the vasculature led to gradual rises in portal pressure until a critical point was reached. An analogy can be drawn to many patients with cirrhosis and portal hypertension who can have subclinical portal hypertensive complications, such as varices, for years until a decompensation point is reached and frequent bleeding ensues.

Dr. Wu: The degree of portal stenosis over time was also unknown, given that we did not have any previous cross-sectional imaging findings. Thus, it is unclear whether the portal stenosis worsened over time in the context of fibrotic remodeling, which could have contributed to a more progressive manifestation of her portal hypertension.

FINAL DIAGNOSIS

Jejunal variceal bleeding.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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